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31N0402

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B.TECH. III SEMESTER MAIN/BACK EXAMINATION 2024-25

COMPUTER SCIENCE AND ENGINEERING

3CS4-02 - Digital Electronics

3AD4-02, 3AM4-02, 3AI4-02, 3CM4-02, 3CY4-02, 3CD4-02, 3CS4-02, 3IT4-02, 3IC4-02 - Common to CS, IT, AI, DS, MC, CM, CD, CA, AD, AM, CY, IO

Time : 3 Hours]

[Max. Marks : 70

[Min. Passing Marks :

Instructions to Candidates :

Part-A : Short Answer Type Questions (up to 25 words) 10 × 2 = 20 marks. All 10 questions are compulsory.

Part-B : Analytical/Problem Solving questions 5 × 4 = 20 marks. Candidates have to answer 5 questions out of 7.

Part-C : Descriptive/Analytical/Problem Solving questions 3 × 10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/ calculated must be stated clearly.

Use of the following supporting materials is permitted during examination. (Mentioned in form no. 205).

1_____

2_____

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Part-A

1. Why the Gray code is it also known as the binary-reflected Gray code ?

2. State De Morgan's theorem.

3. Find the complement of the following expressions :

$$(A + B' + D)\ (A' + B' + C)\ (A + B' + C')$$

4. Prove that an OR gate in a positive logic system is an AND gate in a negative logic system.

5. Briefly describe noise margin of TTL and CMOS logic family.

6. With the help of an example, prove that the dual of the complement of a Boolean expression is the same as the complement of the dual of the same.

7. Convert $(101110)_2$ into Hexadecimal.

8. Differentiate between asynchronous and synchronous counters.

9. Simplify the following SOP equation using K-map technique

$$X = A'B + A'B'C' + ABC' + AB'C'$$

10. Prove that XY + YZ + XZ is a self dual function.

Part-B

1. In which Number system, the following equation is valid :

   (a) $\frac{302}{20} = 12.1$

   (b) $1234 + 4321 = 5555$

   (c) $\sqrt{41} = 5$

2. Use 8 : 1 Mux to implement the following function F and implement F using decoder and NAND gate only :

$$F(A, B, C, D) = \Sigma m\ (1, 2, 4, 5, 9, 10, 13, 14)$$

3. Implement a full adder circuit using 4 : 1 Multiplexers.

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4. With the help of a schematic arrangement, explain how a J-K flip-flop can be used as a (a) a D flip-flop and (b) a T flip-flop.

5. Design a combinational circuit that multiplies by 3 an input decimal digit represented in BCD. The output is also in BCD.

6. Using JK flip-flops, design one typical stage of a register that performs the following logic micro-operations :

$p11 : A \leftarrow (A \vee B)'$ NOR

$p12 : A \leftarrow (A \wedge B)'$ NAND

$p13 : A \leftarrow (A \odot B)'$ Equivalence

7. Explain the operation of TTL NAND gate with a neat circuit diagram. Also discuss the advantage and need of Schottky TTL.

Part-C

1. Design a synchronous counter that counts as 000, 010, 101, 110, 000, 010, ... Ensure that the unused states of 001, 011, 100 and 111 go to 000 on the next clock pulse. Use J-K flip-flops. What will the counter hardware look like if the unused states are to be considered as 'don't care's.

2. Use 7-segment display (as shown in Fig. 1) to show your roll number.

Note : A switch is present in the driver logic which is to be pressed each time to increase the input from 000 to 111.

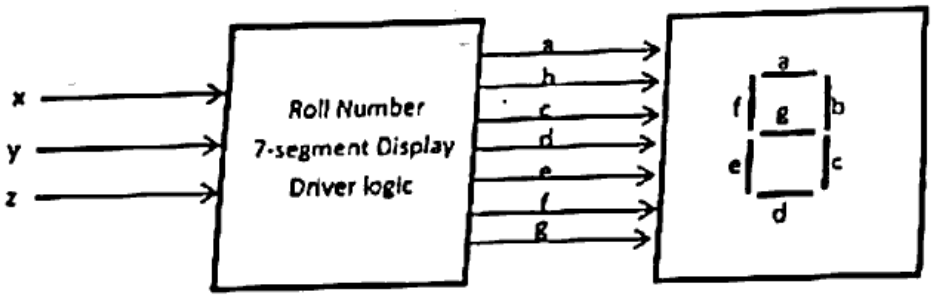


Fig. 1

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3. Using the K-Map method, simplify the following Boolean function and obtain :

(i) minimal SOP and

(ii) minimal POS expression

(iii) implement with 8 × 1 mux

$$F(A, B, C, D) = \Sigma (0, 2, 3, 6, 7) + d(8, 10, 11, 15)$$

(10)

4. Solve the following using Boolean Algebra :

(a) If A'B + AB' = C, Show that :

$$A'C + AC' = B$$

(b) If A'B + CD' = 0, then prove that :

$$AB + C' (A' + D') = AB + BD + B'D' + A'C'D$$

('indicates the compliment of the variable)

5. Using the tabulation method, simplify the following Boolean function :

$F(v, w, x, y, z) = \Sigma(1, 4, 12, 14, 16, 18, 21, 25, 26, 29, 31) + d(0, 2, 5, 30)$

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